

Project Title : AI-Powered Social Media Addiction Detection and Digital Wellbeing Recommendation System

Abstract

Social media platforms have become an important part of modern life and are widely used for communication, entertainment, education, and information sharing. However, excessive and uncontrolled use of social media can lead to problems such as reduced productivity, sleep disturbances, anxiety, stress, loneliness, and poor digital wellbeing. Existing systems often focus mainly on screen-time monitoring and provide limited support for identifying different levels of social media addiction and improving users' digital habits.

This project proposes an AI-Powered Social Media Addiction Detection and Digital Wellbeing Recommendation System (SMADS) that uses machine learning to assess social media usage patterns and predict the addiction level of users. The system uses a social media and mental health dataset containing attributes such as daily screen time, night-time usage, sleep duration, anxiety level, mood score, social comparison, loneliness, physical activity, and life satisfaction. The dataset is preprocessed and relevant features are selected for model development. A Random Forest classification algorithm is used to classify users into Low, Moderate, and High addiction levels and generate an addiction risk score based on their usage and behavioural patterns.

The application is developed using Python, Flask, Pandas, NumPy, Scikit-learn, HTML, CSS, and SQLite. It provides a user-friendly assessment interface, authentication, prediction results, history tracking, progress monitoring, and visual reports to help users understand their social media usage patterns. In addition to addiction detection, the system provides personalized digital wellbeing recommendations, including screen-time reduction strategies, digital detox reminders, healthy sleep suggestions, physical activity recommendations, and progress-based guidance.

The proposed system aims to provide an intelligent and practical approach to identifying social media addiction risk and supporting healthier digital habits. By combining machine learning-based prediction with personalized digital wellbeing recommendations and progress tracking, SMADS helps users become more aware of their social media behaviour and encourages responsible and balanced use of digital platforms.

Keywords

Social Media Addiction, Digital Wellbeing, Machine Learning, Artificial Intelligence, Mental Health, Random Forest, Addiction Risk Prediction.

References

1. Valkenburg, P. M., Meier, A., & Beyens, I. (2022). Social media use and its impact on adolescent mental health: An umbrella review of the evidence. *Current Opinion in Psychology*, 44, 58–68.
2. Plackett, R., Blyth, A., & Schartau, P. (2023). The impact of social media use interventions on mental well-being: Systematic review. *Journal of Medical Internet Research*, 25, e43656.
3. Ye, X. L., Zhang, W., & Zhao, F. F. (2023). Depression and internet addiction among adolescents: A meta-analysis. *Psychiatry Research*, 326, 115311.

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